NUBBLE: A Web-Based Multi-Role Help Desk System for Efficient Task and Inquiry Management

Project Documentation
Presented to the Faculty of the
School of Engineering and Technology – Information Technology
NU Fairview

In Partial Fulfillment
of the Requirements for
Robotic Process Automation (RPASYST)
Database Management 1 (DATAMA1)
UI/UX Design and Programming (USERDES)

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Robotic Process Automation (RPASYST)

I. Introduction

This study introduces **Nubble**, a web-based multi-role help desk system designed to streamline inquiry management within academic environments of not just a specific branch, but to all of its known branches. It aims to provide a centralized, user-friendly platform that allows distinct roles for students, users, staff, and administrators while ensuring efficient communication and inquiry tracking. The system contributes to research by demonstrating a versatile, easy-access, not to one, but to all branches of a university, to seek help from carefully divided departments for organization, and a reliable fast mail-esque inquiry system.

In today's academic scene, efficient communication plays a vital role in giving satisfaction among students that want to enroll, enroll, and their guardians. Traditional methods such as manual logging, sending mails, physically having to be present in a room with the people that can answer your questions, or paper-based inquiries often result in delays, miscommunication, and unorganized records if not managed well. With the growing reliance on digital tools, educational institutions are increasingly shifting toward centralized and automated systems that can manage inquiries, track responses, and maintain accountability across multiple departments and user levels.

The issues addressed by this study are: [1] Students and other users experience slow and uncoordinated inquiry responses due to the lack of a unified digital system. [2] Departments struggle to organize incoming inquiries, leading to inefficiency and missed follow-ups. Sometimes, users need to personally seek staff members in their respective facilities, match their schedules, or communicate with them through platforms that mix academic and non-academic conversations. [3] Existing systems in educational institutions are not fully optimized for role-based access and proper departmental management to effectively filter and categorize questions and inquiries. [4] Communication between university branches remains fragmented, limiting consistent and efficient information flow.

The objectives of the study are: [1] To design and develop a web-based multi-role help desk system that streamlines communication and inquiry management across multiple university branches. [2] To implement role-based access and proper departmental organization for faster, more efficient, and accountable responses. [3] To provide a centralized platform where users, faculty, and staff can interact seamlessly. [4] To enhance institutional responsiveness and transparency through organized digital records and real-time tracking.

The potential contribution of this study is to provide valuable insights and practical advancements in the field of web-based systems and institutional management. It aims to contribute to research by presenting a model for an integrated, multi-role help desk system applicable to academic institutions and organizations. The system is designed to enhance workflow, communication, and task management within departments and branches, supporting institutional digital transformation through a scalable and accessible platform. Furthermore, it promotes efficiency, transparency, and user satisfaction while serving as a reference for future studies and developments in web-based help desk and inquiry management systems.

II. Review Related Literature

Designing a Web-Based Inquiry Management Platform Using a Design Thinking Approach (Case Study: PQM Consultants)

According to **Noor and Ghina (2025)** in **Designing a Web-Based Inquiry Management Platform Using a Design Thinking Approach**, the integration of design thinking to the development of Web-Based Inquiry Management Systems improves the organization operation’s efficiency, responsiveness, and transparency. Their study showed that going from manual processes to web-based platforms helps organization teams manage inquiries more effectively, minimize delays, and communication gaps. It demonstrated how going from manual to digital create more responsive, and collaborative work environments for consulting organizations.

Enhancing accessibility and quality of government services for senior citizens and persons with disabilities

A study from **Cabaddu et al (2025) Enhancing accessibility and quality of government services for senior citizens and persons with disabilities**, explored how help desk platforms is used to assist senior citizens and person with disabilities in Valenzuela City. The study found that these systems improved the efficiency and inclusivity of public services, which allows residents, especially senior citizens and PWD’s, to access support easily. This study demonstrates how the principles of a digital management inquiry system can be applied in community governance to promote faster service for residences.

A Web Based System for Easy and Secured Managing Process of University Accreditation Information

According to **Kommey et al. (2022)**, developing web-based systems for managing accreditation processes in universities helps improve data accuracy, security, and efficiency. Their findings showed that these systems are effective for both handling inquiries and processing and storing academic information. This demonstrates how web-based management tools can support organizational needs.

University of Santo Tomas Department of Filipino Virtual Front Desk

According to Domingo et al. (2021), The Virtual Front Desk introduced at the University of Santo Tomas serves as a digital communication hub that allows students to send inquiries directly to the administrative offices. The system reduces the need for face to face interactions, which simplifies communication throughout different departments. Their findings illustrate how educational institutions are adopting to digital inquiry systems similar to what is described by Noor and Ghina (2025)

Online Support Management System for University of Antique

According to Mission (2020), the implementation of the Online Support Management System within academic institutions serves a more efficient communication between students and administrators. This study highlights that web-based platforms minimise administrative workload while ensuring that students' inquiries are addressed effectively. The findings of this study are consistent with the perspective of Noor and Ghina (2025) where they stated that digital tools enhance responsiveness, efficiency, and transparency in an organization.

WebYu: A Barangay Yuson Web-based Information Management System Initiatives

According to **Tosper Jr et al (2025)** The WebYu Project, a Web-Based Barangay Information Management System for Barangay Yuson, Nueva Ecija, is a system that digitalized resident records and automate document request, significantly reduced manual paperwork's and improved processing items for the Barangay of Yuson. The findings emphasise that web-based platforms can transform local government transactions to be systematic and efficient.

TISSA: A Web-based Helpdesk Support System for Tertiary Institutions with Knowledge-based Management and Ticket Forecasting using Time Series Methods

According to **Arcobel et al (2023)**, A Web Based Helpdesk Support System for Tertiary Institutions was developed to address technical concerns and manage support tickets. The system features a built in knowledge base and ticket tracking functionality that enables faster issue resolution and improved documentation. The study emphasized that digital helpdesk could streamline school operations and make it easier to handle recurring IT problems.

Web-Based Management Information System of Cases Filed with the National Labor Relations Commission

According to **Dela Rosa (2022)**, A Web-Based Management Information System was developed for the National Labor Relations Commissions to enhance the management of case records and inquiries. The study shows how digitalizing the processes made it easier for staff to track case progress, reduced dependency on physical documents, and improved internal coordination and data accessibility. The study emphasized how government offices can greatly benefit from using Web-Based centralized online systems.

Web-based Faculty Development Management System

According to **Badjao et al (2023)**, A Web-Based Faculty Development Management System was developed at Cebu Technological University where it allows multiple users such as faculty members and administrators to access, update, and monitor development records in a web-based centralized platform. The findings emphasized how web-based systems improve accuracy and efficiency in manipulating data. The study also highlighted how readily available and easy to manage systems promote better collaboration.

JuCo-IS: A Development of Web-Based Information System in Judicial Regional Trial Court

According to Acoba et al (2020), JuCo-IS, A Web-Based Information system was designed to to support multi-users, including judicial officers, lawyers, and administrative staffs that allows efficient access, update, and track case records within a unified online system platform. The study found that the use of web-based systems reduced delays and improved data organization among court personnel. The study emphasized how digital systems can improve the Philippine Judicial process through automation.

III. Methodology

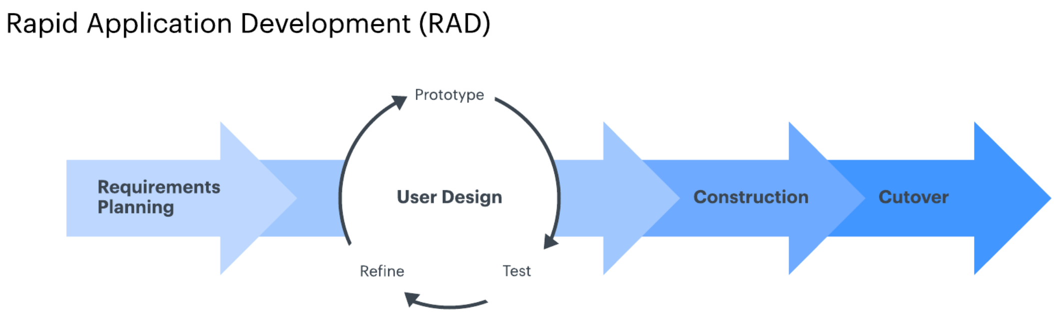

Figure 1.3.1 : Rapid Application Development (RAD)

Introduction to Methodology

The study adopted the Rapid Application Development (RAD) methodology in the development of Nubble: A Web-Based Multi-Role Help Desk System for Efficient Task and Inquiry Management. The RAD approach emphasizes quick and very swift, agile prototyping and adaptive design to achieve faster development cycles while ensuring that user requirements are continuously integrated into the system. This method was selected because of its ability to align closely with user needs and sometimes wants, which is essential for a communication-centered system like Nubble that needs messaging between people to function properly. The approach allows ongoing testing and refinement, ensuring that the system remains efficient and responsive to users' expectations for quick and accurate inquiry management.

Phases of Rapid Application Development

Requirements Planning Phase, data were gathered through informal interviews and observations involving students and from outside the campus. The findings showed that communication gaps were common and that many inquiries were left unresolved or delayed due to the absence of a centralized inquiry platform. Based on this, the objectives of Nubble were established—to create a user-friendly, organized communication system that connects different university roles efficiently through categorized topics and subtopics.

User Design Phase involved the creation of an initial prototype using OutSystems, a low-code development platform suitable for rapid design and testing. The prototype included interfaces for multiple user roles such as normal users, staff, and administrators. Core features included an inquiry submission form, a message interface for Q&A, a dashboard for monitoring inquiry status, and dynamic topic filters. Throughout this phase, Verification Testing was performed, including Unit Testing to confirm individual component functions and Integration Testing to ensure smooth interaction between modules. Continuous feedback from testers led to iterative improvements in responsiveness, data persistence, and navigation clarity.

Construction Phase, the complete Nubble system was implemented. The researcher developed modules for inquiry management, topic and subtopic filtering, real-time tracking, and user authentication with security measures. The database was structured to store inquiries, responses, attachments, and user details according to their assigned roles. Continuous testing during this stage allowed the detection and correction of system errors.

Unit Testing and Integration Testing were repeatedly conducted to verify consistent performance and data flow across all system modules. Tested if client variables worked with local ones properly.

Cutover Phase focused on validation, testing, and deployment. Functional Testing confirmed that essential workflows—such as inquiry submission, replying, and conversation closing—performed as expected. We had random students do a quick test and see if the logic works on the reply and send of questions and answers. And had some personas come and tell us their pain points to us so we could fix it. We tested it on multiple platforms to see if the UI would still look as presentable as it needs to be and if it is indeed compatible with most common platforms and browsers and it is. GTmetrix was also used to test the performance and structure of the system, if it needed improvement, in the overall performance or how the logic nodes were called. We also tested if the data was being saved properly real time and will not be lost when a user refreshes or leaves, or even if they log out of their account thanks to the magic of the client variables, our database integrity was strong and solid, and all data and variables are being passed down properly in the client and server flow.

Database Management 1
(DATAMA1)

I. Business Rules

1. All inquiries must be officially recorded before being processed.
2. Each inquiry must include a clear description of the concern.
3. Inquiries received outside office hours shall be processed on the next working day.
4. Inquiries must be acknowledged within one working day upon receipt.
5. Inquiries must be addressed only through official communication channels.
6. Staff must provide accurate and courteous responses to all inquiries.
7. All responses must be properly documented for record-keeping purposes.
8. Inquiries containing offensive, spam, or irrelevant content will be reported.
9. Resolved inquiries must be marked as “Completed” and archived for future reference.
10. Only authorized personnel may review or respond to client inquiries.
11. Clients must be notified once their inquiry has been resolved.
12. Delayed responses must include a clear explanation and estimated resolution time.
13. All inquiry data shall remain confidential and accessible only to authorized personnel.
14. Staff members are responsible for updating inquiry records after every interaction.
15. Unresolved inquiries exceeding the standard processing time must be escalated to a higher authority.
16. Repeated unresolved or mishandled inquiries may be subject to internal review.
17. Inquiry records must be retained for a set period before being archived or deleted according to policy.

II. Physical Level

A. Entities

1. NUser
2. Branch
3. Topic
4. Inquiry
5. Reports

B. Relationships

1. NUser creates Inquiry
2. NUser employs Branch
3. Branch handles Inquiry
4. Inquiry categorize_as Topics
5. Inquiry generates Reports
6. Reports generated_in Branch
7. Reports submitted_by NUser

III. Logical Level
Entity Relationship Diagram (ERD)

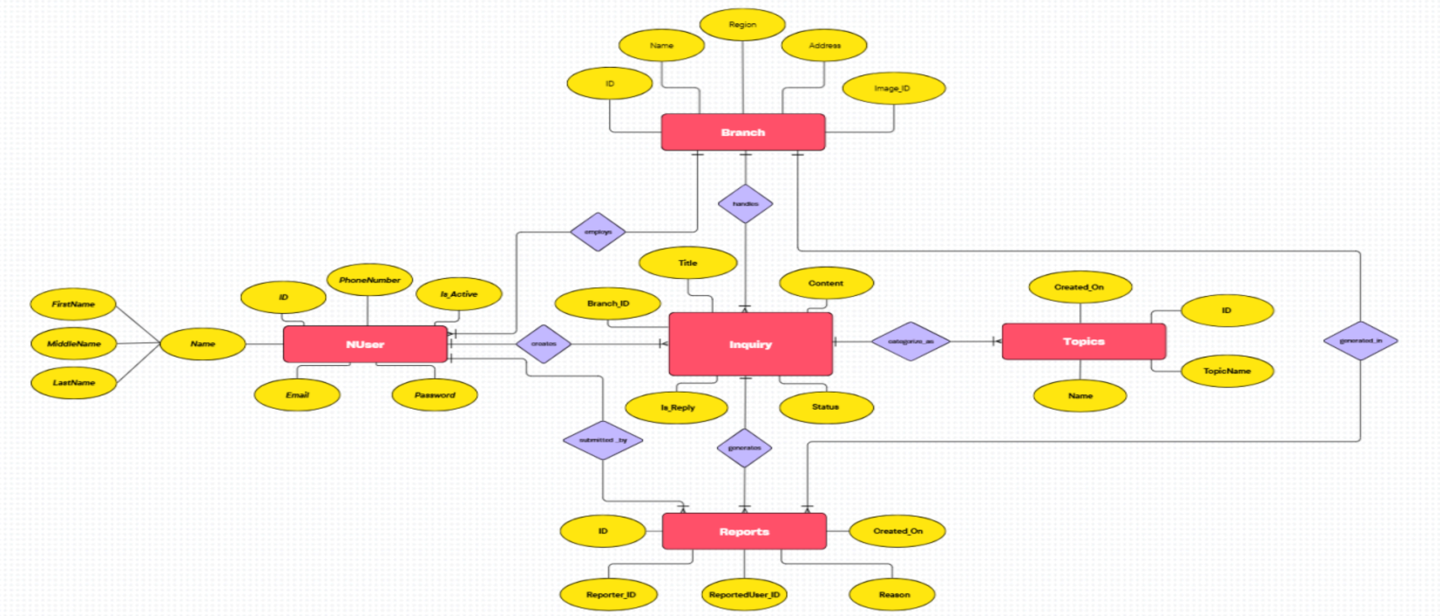

Figure 2.3.1: Entity Relationship Diagram (ERD)

Where as 2 refers to Part 2 of the document which is DATAMA1, 3 refers to Part 3 of the DATAMA1 requirements, and the last number refers to the sequence of images in this part.

IV. View Level

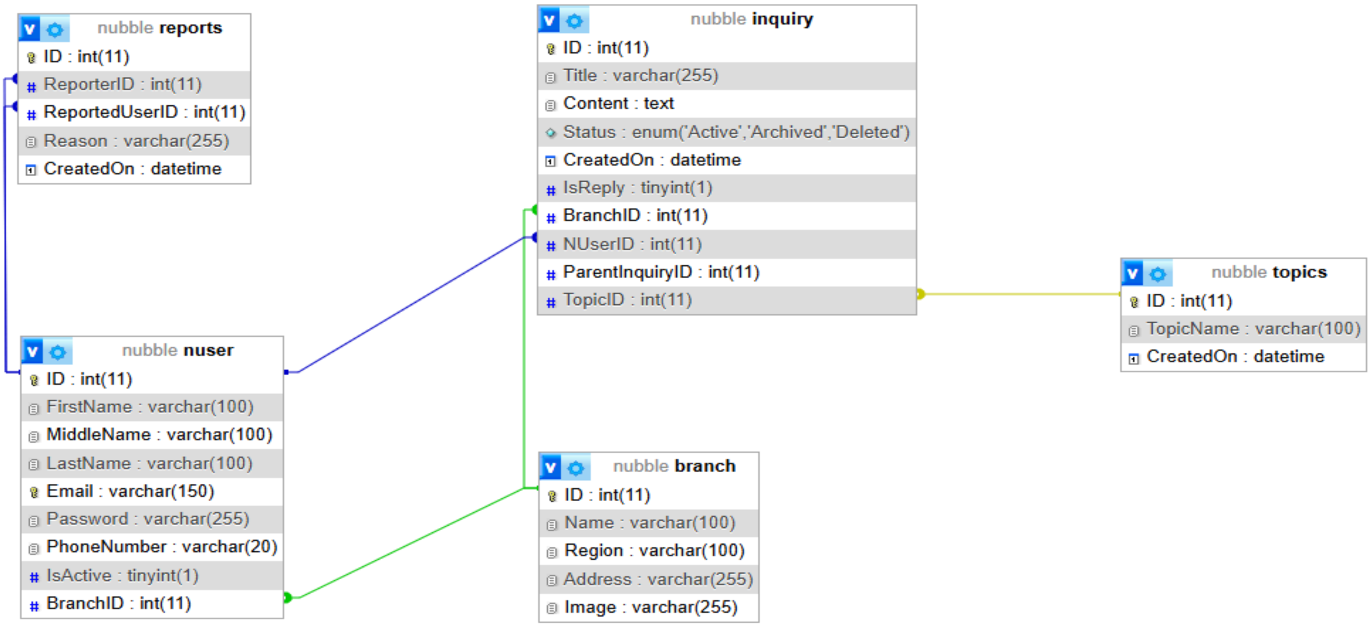

Figure 2.4.1: View Level of the Database from MySQL

Where as 2 refers to Part 2 of the document which is DATAMA1, 4 refers to Part 4 of the DATAMA1 requirements, and the last number refers to the sequence of images in this part.

UI/UX Design and Programming (USERDES)

I. User Personas

MARIA DELA CRUZ
OFFICE SECRETARY

INFO
36 YEARS OLD

San Jose Del Monte, Bulacan

ROLE
A concerned and proactive mother seeking accurate and reliable information about National University branches near her area. She represents parents who are exploring educational opportunities for their children and want to make informed decisions based on convenience, accessibility, and academic quality.

GOALS
- To find the nearest National University branch from their area
- To gather basic information about courses and facilities for her child's potential enrollment
- To easily contact or visit the nearest campus without confusion

PAIN POINTS
- Finds it hard to navigate multiple university websites for location info
- Confused by outdated or incomplete contact details
- Limited free time to make phone inquiries due to work schedule

TECH COMFORT LEVEL
Moderate to High
(She is comfortable using smartphones, search engines, and social media platforms. However, she prefers websites that are straightforward, visually organized, and don't require technical steps or account creation.)

SCENARIO
Maria wants her eldest child to study at National University, but she isn't sure which campus is nearest to San Jose Del Monte. She visits the NU Inquiry Website to search for the closest branch, check available courses, and get directions. She appreciates quick responses and clear navigation to save her time.

Figure 3.1.1: Persona

ERNESTO SANTOS
RETIRED PUBLIC SCHOOL TEACHER

INFO
67 YEARS OLD

Camarin, Caloocan City

ROLE
A retired grandfather who values education and wants to help his family make the best decisions for his grandchild's schooling. He represents senior citizens who often assist younger family members in researching schools, scholarships, and admission requirements. Though not very tech-savvy, he appreciates online platforms that are simple, trustworthy, and easy to navigate, especially when gathering information for loved ones.

GOALS
- To understand step-by-step registration requirements
- To find clear contact information or staff who can assist him

PAIN POINTS
- Struggles with navigating complex online platforms
- Gets easily overwhelmed by too many instructions or pop-ups
- Has difficulty typing or reading small text on screens

TECH COMFORT LEVEL
Low
(He can use a basic smartphone for calls and messaging and occasionally browses Facebook with help from family members. He prefers clear step-by-step instructions and avoids websites that require multiple clicks or form-filling.)

SCENARIO
Ernesto's daughter asked him to help gather information about National University for his grandchild, who will enter college soon. He visits the NU Inquiry Website to check admission requirements, tuition fees, and contact numbers. The site's clean interface and simple navigation make it easy for him to find the details he needs, and he feels satisfied that he can assist his family even with his limited tech experience.

Figure 3.1.2: Persona

Figure 3.1.3: Persona

II. Members' Personas

Figure 3.2.1: Wesly Dela Cruz Persona

Figure 3.2.2: Jimmy II Fortaleza's Persona

Figure 3.2.3: Nedjames Morales Persona

Figure 3.2.4: Matt Mori Tanaka Persona

Figure 3.2.5: Josh Francis Yatco Persona

III. Pain Points

1. Students often don't know where or whom to contact for their concerns.
2. Inquiries take too long to be answered or resolved.
3. Messages and requests sometimes get lost or ignored.
4. People need to visit offices personally just to follow up on simple issues.
5. Users feel frustrated because responses take too long to arrive.
6. Some concerns are repeatedly raised because previous ones were not acknowledged properly.

IV. User Goals

1. To ensure faster and more efficient communication between students, staff, and administrators.
2. To help administrators and staff manage, assign, and resolve inquiries more efficiently.
3. To minimize manual workload by automating the tracking and documentation of inquiries.
4. To reduce delays and prevent inquiries from being ignored, or duplicated.
5. To reduce the need for face-to-face visits by providing a convenient online communication channel.
6. To build user trust by ensuring responsive, transparent, and efficient inquiry handling.
7. To uphold confidentiality and security of all inquiry-related data and user information.

V. Interface

*** Provide here your interfaces with corresponding figure labels below ***
Sample format for figure naming:

Figure 3.5.1: Login Module
Figure 3.5.2: Registration Module

Where as 3 refers to Part 3 of the document which is USERDES, 5 refers to Part 5 of the USERDES requirements, and the last number refers to the sequence of images in this part.